# Almost everywhere monotone additive functions

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**Abstract**

We prove that a real additive function which decreases on a set of natural density zero is a nonnegative constant multiple of the logarithm. This settles a conjecture of Erdős from 1946, listed as Problem 1122 on the Erdős Problems website. The main step is to deduce finite concentration without a growth assumption on the prime values. A truncated normalization and a bounded clipping reduce this step to Mangerel's first-moment theorem for short intervals, Ruzsa's second-moment estimate, and Elliott's fourth-moment inequality.

## 1 Introduction

A function $f\colon \mathbb{N} \to \mathbb{R}$ is *additive* if $f(ab) = f(a) + f(b)$ whenever $(a, b) = 1$. Erdős proved that an additive function which is nondecreasing on $\mathbb{N}$ must be a constant multiple of $\log n$. He conjectured that the same conclusion holds when the set of decreases has natural density zero [1, p. 3]. The conjecture is also recorded as Erdős Problem 1122 [10].

**Theorem 1.1.** *Let $f\colon \mathbb{N} \to \mathbb{R}$ be additive. If*

$$D_f(X) := \#\{n \le X : f(n+1) < f(n)\} = o(X),$$

*then there is a constant $c \ge 0$ such that $f(n) = c \log n$ for every $n \in \mathbb{N}$.*

A Lean 4 formalization of Theorem 1.1 takes the results of Mangerel, Ruzsa, Erdős and Hildebrand as hypotheses.[1]

Erdős proved that $f(n+1) - f(n) = o(1)$ forces $f = c \log$. Kátai obtained the same conclusion from $X^{-1} \sum_{n \le X} |f(n+1) - f(n)| \to 0$ [5]. Wirsing gave another proof under a weaker hypothesis [9]; see also Elliott [3, Ch. 11, pp. 244–247]. Hildebrand proved the density-one version [4], supplying the distribution result Erdős had announced without proof [1, p. 17]. Mangerel obtained an approximate logarithmic representation under the present density hypothesis [6, Theorem 1.9], and the exact conclusion for completely additive functions under additional conditions on prime values and decreases [6, Corollary 1.7].

We say that $f$ has *finite concentration* if there are constants $d > 0$ and $R < \infty$ such that, for arbitrarily large $X$, some interval of length $R$ contains at least $dX$ of the values $f(n)$, $n \le X$. The interval may depend on $X$. Erdős's Theorem V [1] makes finite concentration equivalent to the existence of $c \in \mathbb{R}$ such that

$$\sum_p \frac{\min\{(f(p) - c \log p)^2, 1\}}{p} < \infty. \tag{1.1}$$

[1]The fourth-moment and prime-sum estimates are proved in the formalization; see https://github.com/FireflySentinel/erdos-1122.

Under (1.1), Hildebrand gives a symmetric limiting law for $f(n+1)-f(n)$; the density hypothesis and his corollary then force $f = c\log$. Thus it suffices to prove finite concentration. Assuming the contrary, we normalize $f$ modulo a logarithmic term to fix its truncated quadratic prime mass. Ruzsa gives a positive lower bound for the variance after clipping, while Elliott supplies the fourth-moment control needed to combine Mangerel's short-interval estimate with the density hypothesis and obtain an incompatible upper bound.

### Notation

Letters $p, q$ denote primes. We write

$$\mathbb{E}_X F = \frac{1}{X}\sum_{n\le X} F(n), \qquad \mathcal{A}_H F(n) = \frac{1}{H}\sum_{j=0}^{H-1} F(n-j) \quad (n \ge H).$$

For strongly additive $z(n) = \sum_{p|n} a_p$, put $\mu_z(X) = \sum_{p\le X} a_p/p$. All implicit constants are absolute unless a dependence is indicated. The constants $K$ and $M$ used in the proof are fixed before $X$ tends to infinity. In statements about a family $z_X$, the prime coefficients may depend on $X$.

## 2 Moment estimates and short intervals

For an additive function $z$, define

$$A_0(z,X) = \sum_{p^k\le X} \frac{z(p^k)}{p^k}\left(1-\frac{1}{p}\right), \qquad \widetilde{A}_0(z,X) = \sum_{p^k\le X}\frac{z(p^k)}{p^k},$$

and $B_2(z,X)^2 = \sum_{p^k\le X} |z(p^k)|^2/p^k$. Ruzsa's second-moment estimate [7], in the form recorded by Mangerel [6, Lemma 3.3], gives, for all sufficiently large $X$ and uniformly in $z$,

$$\mathbb{E}_X|z - A_0(z,X)|^2 \asymp \inf_{b\in\mathbb{R}}\left(b^2 + \sum_{p^k\le X}\frac{|z(p^k) - b\log(p^k)|^2}{p^k}\right). \tag{2.1}$$

The upper bound with $b = 0$ is the Turán–Kubilius inequality, valid for all $X \ge 2$ [8, Ch. III.3]. For a strongly additive function $z(n) = \sum_{p|n} a_p$, Elliott's inequality [2, Theorem 1], together with the center estimates below, gives

$$\mathbb{E}_X|z-\mu_z(X)|^4 \ll \left(\sum_{p\le X}\frac{a_p^2}{p}\right)^2 + \sum_{p\le X}\frac{a_p^4}{p}, \qquad X\ge 2. \tag{2.2}$$

If $z$ is strongly additive, then

$$\begin{aligned} |\widetilde{A}_0(z,X) - \mu_z(X)| &\le \sum_{p\le X}\frac{|a_p|}{p(p-1)}, \\ |\widetilde{A}_0(z,X) - A_0(z,X)| &\le \sum_{p^k\le X}\frac{|a_p|}{p^{k+1}}. \end{aligned} \tag{2.3}$$

Both quantities are $\ll (\sum_{p\le X} a_p^2/p)^{1/2}$. Also $B_2(z,X)^2 \le 2\sum_{p\le X} a_p^2/p$. Combining Ruzsa's upper bound with (2.2), if $|a_p|\le L$ and $\sum_{p\le X} a_p^2/p \le V$, then

$$\mathbb{E}_X|z-\mu_z(X)|^2 \ll V, \qquad \mathbb{E}_X|z-\mu_z(X)|^4 \ll V^2 + L^2 V. \tag{2.4}$$

If the coefficients are bounded and $a_p \to 0$ for each fixed prime, dominated convergence in (2.3) gives $A_0(z,X) - \mu_z(X) \to 0$.

**Lemma 2.1.** *Let $z_X$ be strongly additive functions satisfying*

$$|z_X(p)| \le L, \qquad \sum_{p \le X} \frac{z_X(p)^2}{p} \le V$$

*for fixed $L, V < \infty$. Then*

$$\lim_{H\to\infty} \limsup_{X\to\infty} \frac{1}{X} \sum_{H \le n \le X} |\mathcal{A}_H z_X(n) - \mu_{z_X}(X)|^2 = 0, \tag{2.5}$$

*where $H$ runs through the positive integers.*

*Proof.* For $Y \le X$, write

$$d_Y = \frac{2}{Y} \sum_{Y/2 < n \le Y} z_X(n), \qquad F_Y(n) = \mathcal{A}_H z_X(n) - d_Y.$$

Mangerel's Theorem 1.1 [6] gives, for every integer $10 \le H \le Y/100$,

$$\frac{2}{Y} \sum_{Y/2 < n \le Y} |F_Y(n)| \ll \left( \sqrt{\frac{\log\log H}{\log H}} + (\log Y)^{-1/800} \right) \sqrt{2V}. \tag{2.6}$$

Here $B_2(z_X, Y)^2 \le 2V$, and the estimate is uniform over additive functions, so it applies to the family $z_X$.

Counting multiples of each prime gives

$$d_Y - \mu_{z_X}(Y) = \sum_{p \le Y} z_X(p) \left( \frac{2}{Y} \left( \left\lfloor \frac{Y}{p} \right\rfloor - \left\lfloor \frac{Y}{2p} \right\rfloor \right) - \frac{1}{p} \right) = O\left( \frac{L\pi(Y)}{Y} \right). \tag{2.7}$$

Jensen's inequality and (2.4) at $Y$ imply

$$\frac{2}{Y} \sum_{Y/2 < n \le Y} |F_Y(n)|^4 \ll V^2 + L^2 V + O_L((\log Y)^{-4}). \tag{2.8}$$

Indeed, with $\mu = \mu_{z_X}(Y)$, Jensen's inequality gives

$$|\mathcal{A}_H(z_X - \mu)(n)|^4 \le \mathcal{A}_H(|z_X - \mu|^4)(n).$$

Since $H \le Y/100 < Y/2$, each index in these windows lies in $[1, Y]$ and occurs at most $H$ times when we sum over $Y/2 < n \le Y$. Equation (2.7) bounds the center shift. Hölder's inequality now gives

$$\frac{2}{Y} \sum |F_Y|^2 \le \left( \frac{2}{Y} \sum |F_Y| \right)^{2/3} \left( \frac{2}{Y} \sum |F_Y|^4 \right)^{1/3}.$$

Together with (2.6) and (2.8), this proves the required iterated limit on each fixed dyadic band $Y \asymp X$.

To extend this conclusion to the whole interval, fix $\eta = 2^{-r}$. On each of the $r$ dyadic bands covering $(\eta X, X]$, we have

$$|\mu_{z_X}(X) - \mu_{z_X}(Y)| \le L \sum_{Y < p \le X} \frac{1}{p} = o(1).$$

The omitted initial segment $H \le n \le \eta X$ contributes $O_{L,V}(\sqrt{\eta})$ to (2.5): apply Cauchy–Schwarz and Jensen, using the fourth-moment bound (2.4) at $X$. Given $\varepsilon > 0$, first choose $\eta$ so that this bound is less than $\varepsilon/2$. The band estimates then give $H_0$ such that, for every integer $H \ge H_0$, their combined contribution is less than $\varepsilon/2$ for all $X \ge X_0(\varepsilon, \eta, H)$. This proves (2.5). □

## 3 A weighted estimate for a set of primes

**Lemma 3.1.** *Suppose $0 < M < 1/4$ and $\mathcal{T}$ is a set of primes at most $X$ such that $\sum_{p\in\mathcal{T}} 1/p \le M$. Then, uniformly in $\mathcal{T}$,*

$$\sum_{p\in\mathcal{T}} \frac{1}{p} \sum_{X/p<q\le X} \frac{1}{q} \ll M\log(2/M) + o_{X\to\infty}(1). \tag{3.1}$$

*Here $M$ is fixed in the term $o(1)$.*

*Proof.* We use Mertens's prime harmonic-sum formula and the standard prime-counting bound; see [8]. For $p \le X^{1-M}$, the inner sum is at most $\log(1/M) + o(1)$. These primes contribute at most $M\log(1/M) + o(1)$.

For $p > X^{1-M}$, discard the restriction $p \in \mathcal{T}$ and interchange the sums. The terms with $q > X^M$ are bounded by

$$\left(\sum_{X^{1-M}<p\le X} \frac{1}{p}\right)\left(\sum_{X^M<q\le X} \frac{1}{q}\right) \ll M\log(2/M) + o(1).$$

For $q \le X^M$, the condition $X/p < q$ implies $p > X/q$. The bound $\pi(t) \ll t/\log t$ and partial summation give, uniformly for $2 \le q \le X^M$,

$$\sum_{X/q<p\le X} \frac{1}{p} \ll \frac{\log q}{\log X}.$$

Their total is $\ll (\log X)^{-1} \sum_{q\le X^M} (\log q)/q \ll M$, proving (3.1). □

## 4 Normalization

In the remainder of the proof, suppose that $f$ does not have finite concentration. For $s > 0$ define

$$W_X(s) = \min_{c\in\mathbb{R}} \left\{ c^2 + \sum_{p\le X} \frac{\min\{(f(p)/s - c\log p)^2, 1\}}{p} \right\}. \tag{4.1}$$

**Lemma 4.1.** *For each fixed $M > 0$ and all sufficiently large $X$, there are $s_X > 0$ and a minimizer $c_X$ in (4.1) such that $W_X(s_X) = M$. Then necessarily $s_X \to \infty$, and any corresponding minimizers satisfy $c_X \to 0$. Put $u_p = f(p)/s_X - c_X \log p$ for $p \le X$. Then*

$$\sum_{p\le X} \frac{\min\{u_p^2, 1\}}{p} = M - c_X^2 = M - o(1), \tag{4.2}$$

$$\sum_{\substack{p\le X\\ |u_p|<1}} \frac{u_p \log p}{p} = c_X. \tag{4.3}$$

*No $u_p$ has absolute value exactly $1$.*

*Proof.* The expression in braces in (4.1) is continuous and coercive as a function of $c$, so its minimum is attained. Every minimizer satisfies $c^2 \le \sum_{p\le X} 1/p$, by comparison with $c = 0$. The minimum can therefore be taken over a fixed compact interval, independent of $s$, and is continuous in $s > 0$. On writing $\lambda = sc$, the expression becomes

$$\frac{\lambda^2}{s^2} + \sum_{p \le X} \frac{\min\{(f(p) - \lambda \log p)^2/s^2, 1\}}{p}.$$

It follows that $W_X(s)$ is nonincreasing in $s$, and it tends to zero as $s \to \infty$ for fixed $X$.

For each fixed $s > 0$, $W_X(s) \to \infty$. Otherwise some unbounded sequence of $X$ would have bounded $W_X(s)$ and bounded minimizers. Pass to a subsequence along which the minimizers converge to $c$. Fatou's lemma gives

$$\sum_p \frac{\min\{(f(p)/s - c \log p)^2, 1\}}{p} < \infty.$$

Rescaling by the fixed number $s$ preserves convergence of the truncated square sum. Erdős's Theorem V would therefore give finite concentration of $f$, contrary to the assumption.

Continuity now supplies $s_X$ with $W_X(s_X) = M$. Monotonicity and the preceding claim imply $s_X \to \infty$ for every such choice. We have $|c_X| \le \sqrt{M}$. If a subsequential limit of $c_X$ were a nonzero number $c$, Fatou's lemma on the fixed primes would give

$$\sum_p \frac{\min\{(c \log p)^2, 1\}}{p} \le M,$$

which is impossible. Thus $c_X \to 0$, and (4.2) follows.

It remains to prove the stationary identity and exclude equality at the truncation threshold. Let $G(c)$ denote the expression in braces in (4.1), with $s = s_X$. For any subset $J$ of the primes at most $X$, the quadratic

$$Q_J(c) = c^2 + \sum_{p \in J} \frac{(f(p)/s_X - c \log p)^2}{p} + \sum_{\substack{p \le X \\ p \notin J}} \frac{1}{p}$$

lies above $G(c)$. Choose $J = \{p \le X : |u_p| < 1\}$. Then $Q_J(c_X) = G(c_X)$, so $c_X$ also minimizes $Q_J$. The equation $Q_J'(c_X) = 0$ is exactly (4.3). If $|u_p| = 1$ for some $p$, the quadratic $Q_{J \cup \{p\}}$ also touches $G$ at $c_X$ and must have derivative zero there, contradicting

$$Q'_{J \cup \{p\}}(c_X) - Q'_J(c_X) = -\frac{2u_p \log p}{p} \ne 0. \qquad \square$$

# 5 Clipping the normalized function

Fix $K > 1$ and $0 < M < 1/4$, and use Lemma 4.1. Let $\phi_K(t) = \max\{-K, \min\{t, K\}\}$ and set

$$\mathcal{T} = \{p \le X : |u_p| > 1\}, \qquad S_0(n) = \sum_{\substack{p \mid n \\ p \notin \mathcal{T}}} u_p,$$

$$a_X = \sum_{\substack{p \le X \\ p \notin \mathcal{T}}} \frac{u_p}{p}, \qquad S(n) = S_0(n) - a_X.$$

Coefficients at primes greater than $X$ are set to zero. Define

$$\begin{aligned} z_X(n) &= \sum_{p|n} \phi_K(u_p), \qquad Z(n) = z_X(n) - a_X, \\ g_X(n) &= \phi_K\left(\frac{f(n)}{s_X} - c_X \log n - a_X\right). \end{aligned} \tag{5.1}$$

Thus $g_X$ clips the value of the additive function, whereas $z_X$ clips the prime contributions individually.

**Lemma 5.1.** *There is an absolute constant $C$ such that*

$$\limsup_{X\to\infty} \mathbb{E}_X|g_X - Z|^2 \le C\left(\frac{M}{K^2} + M^2\log(2/M) + K^2M^2\right). \tag{5.2}$$

*Proof.* The small coefficients have quadratic mass at most $M$, and $\sum_{p\in\mathcal{T}} 1/p \le M$. Hence

$$\mathbb{E}_X S^4 \ll M + M^2. \tag{5.3}$$

Put $T(n) = \#\{p \in \mathcal{T} : p \mid n\}$. Counting multiples of distinct primes gives

$$\mathbb{E}_X T(T-1) = \sum_{\substack{p,q\in\mathcal{T}\\ p\ne q}} \frac{1}{X}\left\lfloor \frac{X}{pq}\right\rfloor \le M^2. \tag{5.4}$$

Moreover,

$$\mathbb{E}_X(S^2\mathbf{1}_{T\ge1}) \ll M^2\log(2/M) + o(1). \tag{5.5}$$

For $p \in \mathcal{T}$, the coefficient of $p$ in $S_0$ is zero, so $S_0(pm) = S_0(m)$ for every $m$, including multiples of $p$. Writing $y = X/p$, for $y \ge 2$ equation (2.4) gives

$$\begin{aligned} \mathbb{E}_X(S^2\mathbf{1}_{p|n}) &= \frac{1}{p}\frac{1}{y}\sum_{m\le y}|S_0(m) - a_X|^2 \\ &\ll \frac{1}{p}\left(M + |a_X - \mu_{S_0}(y)|^2\right). \end{aligned}$$

For $1 \le y < 2$, the same bound follows directly from $S_0(1) = \mu_{S_0}(y) = 0$. By Cauchy–Schwarz,

$$|a_X - \mu_{S_0}(X/p)|^2 \le M \sum_{X/p<q\le X} \frac{1}{q}.$$

Hence, using $\mathbf{1}_{T\ge1} \le \sum_{p\in\mathcal{T}} \mathbf{1}_{p|n}$,

$$\begin{aligned} \mathbb{E}_X(S^2\mathbf{1}_{T\ge1}) &\ll M\sum_{p\in\mathcal{T}}\frac{1}{p} + M\sum_{p\in\mathcal{T}}\frac{1}{p}\sum_{X/p<q\le X}\frac{1}{q} \\ &\ll M^2 + M^2\log(2/M) + o(1), \end{aligned}$$

by Lemma 3.1. This proves (5.5).

Set

$$g_X^*(n) = \phi_K\left(S(n) + \sum_{\substack{p|n\\ p\in\mathcal{T}}} u_p\right).$$

If $T(n)=0$, the difference $g_X^*(n)-Z(n)$ vanishes for $|S(n)|\le K$ and otherwise has absolute value at most $|S(n)|$. Its squared contribution is at most $\mathbb{E}_X S^4/K^2$. If $T(n)=1$, with unique tail value $v$, the Lipschitz property of $\phi_K$ gives

$$|\phi_K(S+v)-S-\phi_K(v)|\le 2|S|.$$

If $T(n)\ge 2$, then

$$|g_X^*(n)-Z(n)|\le |S(n)|+K(T(n)+1)\le |S(n)|+\frac{3}{2}KT(n),$$

and hence

$$|g_X^*(n)-Z(n)|^2\le CS(n)^2+CK^2T(n)(T(n)-1).$$

Equations (5.3)–(5.5) therefore imply

$$\mathbb{E}_X|g_X^*-Z|^2\ll \frac{M}{K^2}+M^2\log(2/M)+K^2M^2+o(1). \tag{5.6}$$

To compare $g_X$ and $g_X^*$, suppose $p^k\|n$ with $k\ge 2$. Its contribution to the difference between the two unclipped expressions is

$$\frac{f(p^k)-f(p)}{s_X}-c_X(k-1)\log p.$$

This tends to zero for each fixed prime power. Moreover, $\sum_{p,k\ge 2}p^{-k}<\infty$. Given $\varepsilon>0$, choose a finite set $\mathcal{F}$ of prime powers such that $\sum_{p^k\notin\mathcal{F},\,k\ge 2}p^{-k}<\varepsilon$. Then, uniformly in $X$,

$$\frac{1}{X}\#\{n\le X:\ \text{some } p^k\notin\mathcal{F},\ k\ge 2,\ \text{divides } n\}<\varepsilon.$$

Outside this exceptional set, only the fixed prime powers in $\mathcal{F}$ contribute, and their total contribution tends uniformly to zero. The Lipschitz bound and $|g_X|,|g_X^*|\le K$ therefore give

$$\limsup_{X\to\infty}\mathbb{E}_X|g_X-g_X^*|^2\le 4K^2\varepsilon.$$

Letting $\varepsilon\downarrow 0$ yields $\mathbb{E}_X|g_X-g_X^*|^2=o(1)$; combine this with (5.6). □

**Lemma 5.2.** *There are absolute constants $c_0,C_0>0$ such that*

$$\liminf_{X\to\infty}\mathbb{E}_X|z_X-\mu_{z_X}(X)|^2\ge c_0M-C_0K^2M^2. \tag{5.7}$$

*Proof.* Write $v_p=\phi_K(u_p)$. Since $K>1$, (4.2) implies

$$M-o(1)\le\sum_{p\le X}\frac{v_p^2}{p}\le K^2M. \tag{5.8}$$

On $p\notin\mathcal{T}$ we have $v_p=u_p$, so (4.3) gives

$$\left|\sum_{p\le X}\frac{v_p\log p}{p}\right|\le |c_X|+KM\log X. \tag{5.9}$$

By partial summation from Mertens's first theorem, $\sum_{p\le X}(\log p)^2/p \sim \frac{1}{2}(\log X)^2$. The quadratic projection identity gives

$$\begin{aligned}\inf_{b\in\mathbb{R}} \sum_{p\le X} \frac{(v_p - b\log p)^2}{p} &= \sum_{p\le X}\frac{v_p^2}{p} - \frac{\left(\sum_{p\le X} v_p \log p/p\right)^2}{\sum_{p\le X}(\log p)^2/p} \\ &\ge M - CK^2M^2 - o(1). \end{aligned} \tag{5.10}$$

Apply (2.1), dropping the nonnegative $b^2$ term and the terms with $k \ge 2$ from its right-hand side. For each fixed prime $p$, $v_p \to 0$, because $s_X \to \infty$ and $c_X \to 0$. Therefore (2.3) and dominated convergence give $A_0(z_X, X) - \mu_{z_X}(X) \to 0$. The second moments are uniformly bounded by (2.4) and (5.8); changing the center consequently changes the second moment by $o(1)$. Combining this with (5.10) proves (5.7). $\square$

# 6 Proof of Theorem 1.1

Assume that $f$ has no finite concentration and take $g_X$ from (5.1). If $f(n+1) \ge f(n)$, monotonicity and Lipschitz continuity of $\phi_K$ give $(g_X(n)-g_X(n+1))_+ \le |c_X|\log(1+1/n)$, where $t_+ = \max\{t, 0\}$. At a decrease of $f$, this quantity is at most $2K$. Since $|g_X| \le K$, telescoping the signed increments gives

$$\sum_{n<X} |g_X(n+1) - g_X(n)| \le 2K + 4KD_f(X) + 2|c_X|\log X = o(X). \tag{6.1}$$

For each fixed $H$, every difference $g_X(n) - g_X(n-j)$ is bounded by the sum of the $j$ intervening absolute increments. Since $|g_X| \le K$, (6.1) yields

$$\frac{1}{X}\sum_{H\le n\le X} |g_X(n) - \mathcal{A}_H g_X(n)|^2 \longrightarrow 0. \tag{6.2}$$

Let $m_X = \mu_{z_X}(X) - a_X$. Then $Z - m_X = z_X - \mu_{z_X}(X)$, and

$$Z - m_X = (Z - g_X) + (g_X - \mathcal{A}_H g_X) + \mathcal{A}_H(g_X - Z) + (\mathcal{A}_H Z - m_X). \tag{6.3}$$

The square of the right-hand side is at most four times the sum of the four squared terms. Jensen's inequality gives the contraction

$$\frac{1}{X}\sum_{H\le n\le X} |\mathcal{A}_H(g_X - Z)(n)|^2 \le \frac{1}{X}\sum_{n\le X}|g_X(n) - Z(n)|^2.$$

Lemma 2.1 applies to $z_X$, whose prime coefficients are bounded by $K$ with quadratic mass at most $K^2M$. The first $H$ indices contribute $o(1)$ by (2.4) and Cauchy–Schwarz. Taking $X \to \infty$ and then $H \to \infty$ in (6.3), and using (6.2), gives

$$\begin{aligned}\limsup_X \mathbb{E}_X|z_X - \mu_{z_X}(X)|^2 &\le 8\limsup_X \mathbb{E}_X|g_X - Z|^2 \\ &\le C_1\left(\frac{M}{K^2} + M^2\log(2/M) + K^2M^2\right)\end{aligned} \tag{6.4}$$

for an absolute constant $C_1$.

Choose $K > 1$ so large that $C_1/K^2 < c_0/4$, where $c_0$ is from Lemma 5.2. Next choose $0 < M < 1/4$ so small that

$$C_1 M\log(2/M) + (C_1 + C_0)K^2M < c_0/4.$$

With these fixed choices, (6.4) contradicts (5.7). Hence $f$ has finite concentration.

Erdős's Theorem V now gives (1.1). Hildebrand's Theorem 1 gives a limiting distribution for $f(n+1) - f(n)$ whose characteristic function [4, (1.4)] is a product of real factors, hence the law is symmetric. For every $\varepsilon > 0$, $\#\{n \le X : f(n+1) - f(n) < -\varepsilon\} \le D_f(X) = o(X)$. Since this holds for every $\varepsilon > 0$, the limiting law assigns no mass to $(-\infty, 0)$; by symmetry it is therefore $\delta_0$. Hence $f(n+1) - f(n) \to 0$ in density, and a standard diagonal argument yields a density-one set along which the differences tend to zero. Hildebrand's corollary [4, pp. 258–259] now gives $f(n) = c \log n$ for some real $c$ and all $n$. Finally, $c < 0$ would give a decrease at every integer, so $c \ge 0$. $\square$

## Declaration of generative AI use

GPT-6 Astra generated proofs and manuscript drafts, GPT-5.6 Sol and Claude Opus 5 assisted with editorial review, and OpenAI Codex (GPT-6) generated the Lean code. The Lean 4 development checks the implication from the stated analytic hypotheses to Theorem 1.1, without formalizing the cited theorems.

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